

sew:bit

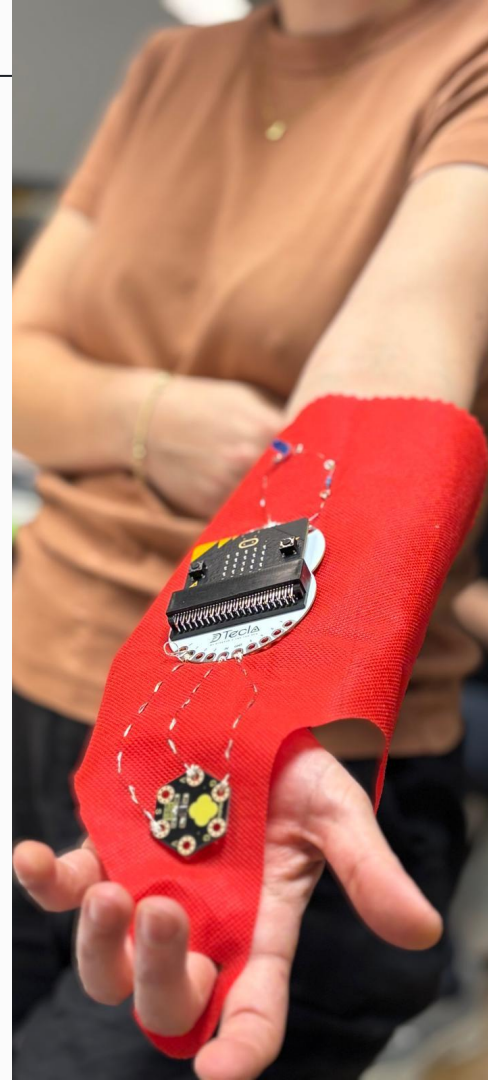
An Open-Source Expansion Board Extending the micro:bit for E-Textile and Wearable Computing in K-6 Education

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Leirado, Domínguez & Fuentemilla (2026) — Design case study



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1. Purpose of the paper

A widely used platform, an underserved valuable practice

11.5 M

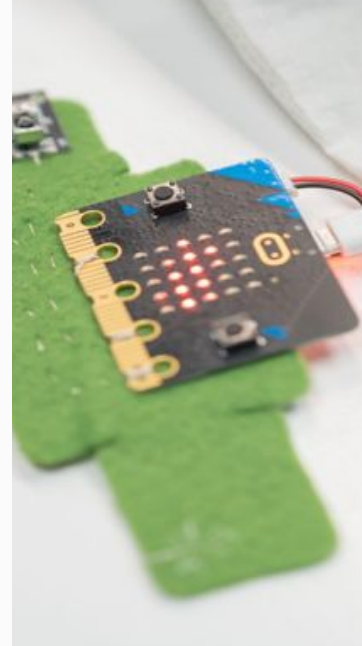
micro:bit devices
distributed by 2025

~70 M

Young people
reached worldwide

Scarce

Hardware adaptation
for e-textiles



E-textiles are a powerful route to equitable STEAM, engaging diverse learners — especially girls — by blending craft with code and circuitry (Buechley 2008; Rode 2015).

Yet the **BBC micro:bit — the most adopted K-6 physical-computing platform** — lacks hardware designed for textile-based making. LilyPad, Flora and Circuit Playground shaped wearable computing, but none adapt the micro:bit for the classroom stock schools already own.

1. Purpose of the paper

No optimal way to adapt the micro:bit for e-textiles

Option	Pros	Cons
Direct sewing of exposed connectors	No additional components needed	Limited to 3 pins Board becomes permanent (must cut stitches to repurpose)
Crocodile clips (Kitronik)	Reusable: connect/disconnect BBC micro:bit without cutting stitches	Limited to 3 pins Narrow holes incompatible with large beginner sewing needles
SewSimple (Feng et al., 2025)	Flexible fabric extension for wearable integration Large through-hole (TH) pads for beginner needles	Limited to 3 pins Not commercially available (Research)
Keystudio Honeycomb kit	Edge connector for reusability 10 pins for sensors/actuators Wide sensor/actuator range	Bottom connectors cause forward tilt (BBC micro:bit weight hinders wearables) Some sensors redundant or too advanced for K-9 Missing beginner-friendly sensors

1. Purpose of the paper

Contribution of the paper



01 Sew:bit board

— An open-source expansion board that adapts the micro:bit for textile-based making.

02 Documented design case

— Iterative development within a Fab Lab context: decisions, failures, fixes.

03 Early findings

— From deployments with K-6 students, teachers and makers on usability, accessibility and pedagogy.

2. Design/ methodology/ approach

A design case study, not a controlled evaluation

Iterative cycles of **design** → **fabrication** → **workshop** → **reflection**.

Iteration, repair and adaptation are treated as sources of design knowledge (Boling, 2010).

3 full cycles over 1 year, each: **problem analysis** • **prototyping** • **workshop enactment** • **redesign**.

— Computational making

Integrates processes of fabrication and programming (Rode et al, 2015)

— E-textiles pedagogy

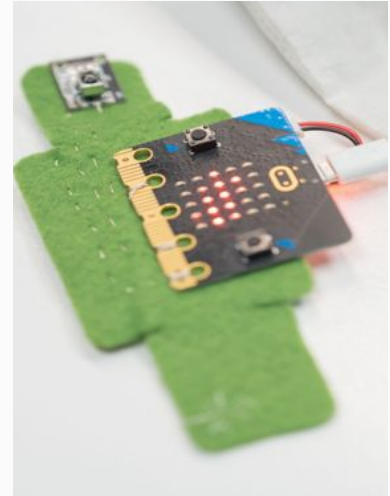
Equitable STEAM through wearable tech (Buechley 2008)

— Fablabable open hardware

Locally replicable designs for education

Design priorities

- micro:bit ecosystem compatibility
- Ease of introduction for non-experts
- Robustness for classroom use
- Affordability

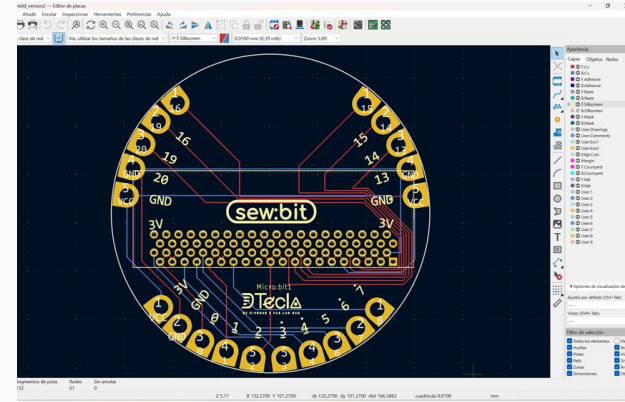
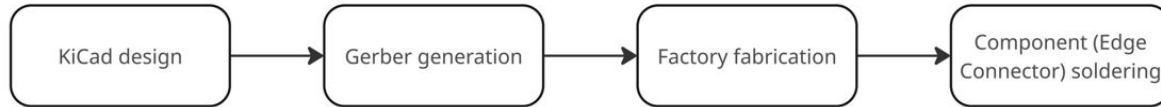


2. Design/ methodology/ approach

Cycle 1: basic shield – Form factor first : a circular, sewable shield

- **Aesthetics & accessibility** — craft look (à la LilyPad); easy edge-connector insert/removal for multi-project reuse.
- **Physical layout** — circular shape → traces radiate from centre, spacing pads and reducing shorts.
- **Sewability** — oversized pads for large-hole needles; crocodile-clip friendly.

Design - fabrication workflow:



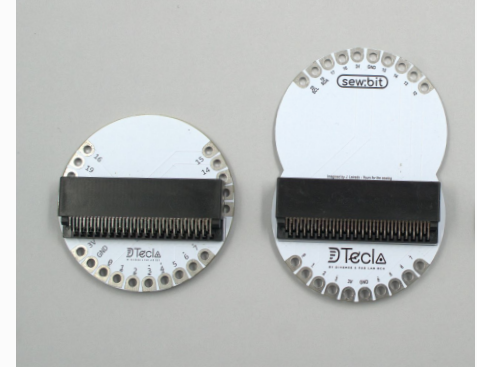
Workshop findings:

- TH edge connector too fine to mill locally → outsourced.
- Footprint error,
- pin occlusion,
- unstable on fabric.
- 10 prototypes.

2. Design/ methodology/ approach

Cycle 2: two intersecting circles, top/bottom pins

- **Thinner board** 1.6 → 1 mm (−37.5%) for a lighter feel.
- **New shape** from two intersecting circles — exposes all pins, keeps rounded aesthetics.
- **Top/bottom split pins** anchor stitching and stop forward tilt.
- **Side gap** holds a LiPo battery by geometry alone.



2 workshop enactment

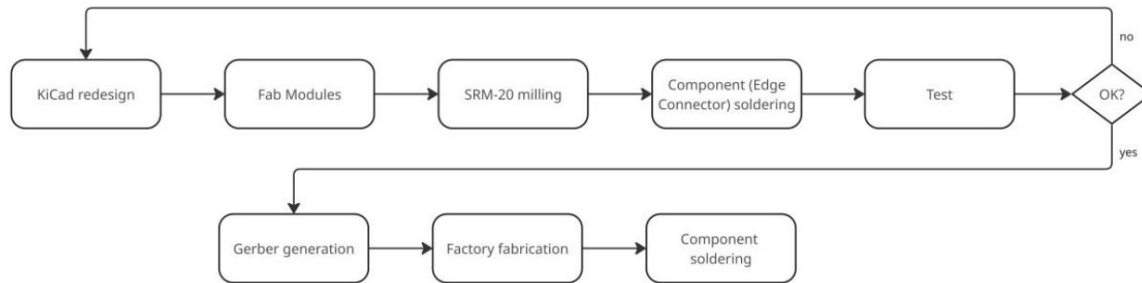
- 12 K-6 teachers (DTecla) — 2-day sound e-textile; sew:bit + Keyestudio sensors eased use, but many sensors were redundant with the micro:bit.
- 12 kids (Maker Faire Galicia) — laser-tag; bent-leg IR sensors were hard to connect, and most had never sewn.

2. Design/ methodology/ approach

Cycle 3: Locally fabricable, with a sensor kit + perforated canvas

- **SMD edge connector** — millable on a Roland SRM-20 → full in-house fabrication, smaller board.
- **Sewable sensor kit** — tackles sewing frustration, uses the 17 exposed pins (LDR, tilt, IR, MOSFET).
- **Pre-perforated canvas** — laser-cut, scaffolds beginner sewing.

Extended design - fabrication



FAB25 - Brno

15 makers built e-textile aprons.
Perforated fabric + sewable sensors
were well received.

3. Data sample

Workshop contexts as sources of design reflection

No formal data-collection study. Observations, informal conversations and workshop notes were used descriptively to inform each redesign.

Cycle	Pros	Participants	Workshop purpose
Cycle 1	Fab Lab Bcn / Fundació Diverse staff	3	Internal prototype testing & design refinement
Cycle 2.1	K-6 teachers (Dtecla)	12	Explore classroom-oriented e-textiles
Cycle 2.2	Children (Maker Faire Galicia)	12	Explore beginner use of the prototype
Cycle 3	Maker community (FAB25)	15	Explore fablable sensor & board use

4. Findings

Four recurring design insights

01	02	03	04
Hardware usability & sewability	Learning & engagement	Classroom integration & scaffolding	Fab Lab reception & replication
By Cycle 3, pre-perforated fabric + dedicated sewable sensors reduced shorts — reliable circuits within 1.5-hour workshops.	Teachers felt confident to reuse the board; kids personalised laser-tag wearables; makers built diverse apron interactions.	Teachers valued micro:bit compatibility but flagged time limits, need for pre-packaged kits and basic sewing gaps.	Fab25 makers responded positively and asked for full docs: KiCad/Gerber/BOM, lesson plans, parametric files.

5. Research limitations

Situated insights, not generalizable results

Bounds of this work

- Workshops sat within connected Fab Lab / maker networks, not independent schools.
- Modest number of enactments — no controlled comparison or statistical generalization.
- Prototype stage: not studied at manufacturing scale or in longitudinal classroom use.

Where to go next

- Deploy across a broader, more diverse range of Fab Labs and schools.
- Formal classroom studies with a clear methodology.
- Comparative testing against commercial kits.
- Longitudinal use + reflections from teachers, makers and children.

6. Practical implications

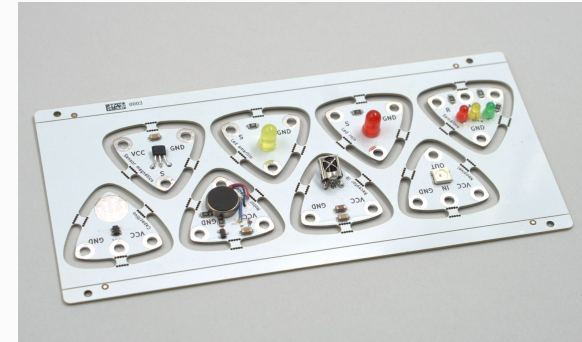
What sew:bit offers each audience

- | | | | |
|-----------|--------------------------------|-------|--|
| 01 | Fablab & makerspaces | ————— | SMD connector → full in-house milling of double-sided boards. Download the complete repo (KiCad, Gerber, BOM, assembly) and adapt to regional parts. |
| 02 | K-6 classroom teachers | ————— | Extends existing micro:bit inventory — minimal new purchases. A learning journey from perforated-fabric sensor projects to curriculum-integrated work. |
| 03 | micro:bit e-textiles designers | ————— | Four transferable design principles to guide future wearable hardware for the micro:bit. |

6. Practical implications

What we'd tell the next hardware designer

- **Top/bottom pin distribution:** Prevents forward tilt; keeps natural fabric drape when sewn.
- **Oversized sewable pads (≥ 3 mm):** Fit large-hole beginner needles and crocodile clips for prototyping.
- **Millable SMD edge connector:** Enables local Fab Lab production (Roland SRM-20 compatible).
- **Modular sensor kits:** Ready-to-sew sensors optimised for pedagogical projects; less improvisation.



6. Practical implications

A fable micro:bit e-textile ecosystem

Novel elements

- Three documented Fab Lab iterations — failures (pin occlusion, forward tilt, sensor leg-bending) and their fixes, with full replication files.
- Four micro:bit-specific design principles for e-textile hardware.
- A K-6 pedagogical pathway from scaffolded sensor exploration to curriculum projects — addressing a recognised research gap.



8. Takeaways

sew:bit brings e-textiles to the world's most deployed classroom microcontroller: openly, and fabricable anywhere.

Open by design KiCad • Gerber • BOM • assembly guides • MakeCode • parametric files — for local adaptation and replication.

Acknowledge

Developed within the DTecLa project — Fundació Diverse × Fab Lab Barcelona — supporting STEAM/maker education in vulnerable school environments. Thanks to the children, teachers and makers across the three workshops, and to Fab Lab Barcelona staff (Josep, Dani, Adai).